

Problem Statement: Bitcoin Price Prediction using Machine Learning

The Challenge:

Bitcoin is a highly volatile cryptocurrency. Predicting its price movement is critical for investors, traders, and policymakers to mitigate risk, optimize portfolios, and make informed financial decisions. Due to the high noise and non-linear patterns in financial time-series data, traditional forecasting methods are highly inaccurate.

The Objective:

Design, build, and evaluate a machine learning regression model that can accurately forecast the Close price of Bitcoin based on historical market indicators (Open, High, Low, Volume, and time-series lag/rolling features). The goal is to build a reliable predictive model that can assist in trading decisions and risk assessment.

Technology Stack:

- IBM Cloud & watsonx.ai Studio: For pipeline execution and runtime hosting.
- AutoAI / Granite Models: For automated feature engineering and hyperparameter tuning.
- Scikit-Learn Regression Models: Evaluated pipelines including Random Forest Regressor and Gradient Boosting Regressor.

Author Details:

Student Name: Raj Kumar Gupta

College: Amity University Jharkhand

Email: rajkrgupta8757299494@gmail.com

AICTE ID: STU6843fe22befb11749286434